Rational Normalization of Local Valuation Tests

10 November 2026

Abstract

We give rational functions which suppress local conditions at a prime where an auxiliary coefficient has odd valuation, without requiring any bound on the poles of the original functions. At a prime where that coefficient is a unit, positive valuation of the normalized function is equivalent to positive valuation of the original one. A second result uses two balanced contact functions with disjoint contact and pole sets: after a suitable change of scale, at least one is a unit at each prime of odd valuation of the scalar coefficient. Together these results supply local extraction statements for products of two arithmetic coefficients. The statements concern local conditions and do not by themselves establish a smaller defining polynomial for the integers.

1 A rational function with controlled valuations

Let L be a finite extension of $\mathbb{Q}_p$ unramified over $\mathbb{Q}_p$. Normalize its valuation by $v(p) = 1$, so that $v(L^\times) = \mathbb{Z}$, and put $v(0) = +\infty$. For $z \in L^\times$, define

$$\Psi_z(D) = \frac{zD^2}{1 + zD^2}$$

whenever the denominator is nonzero. All assertions involving this rational function include that domain condition.

Lemma 1.1. *The following statements hold.*

1. *If $v(z)$ is odd, then the denominator of $\Psi_z(D)$ is nonzero for every $D \in L$, and*

$$v(\Psi_z(D)) = \max\{0, v(z) + 2v(D)\} \geq 0 \qquad (D \neq 0).$$

   For $D = 0$ the value of the rational function is zero.

2. *If $v(z) = 0$, then*

$$v(\Psi_z(D)) > 0 \quad \Longleftrightarrow \quad v(D) > 0.$$

   When $0 < v(D) < +\infty$, one has $v(\Psi_z(D)) = 2v(D)$.

Proof. For $D \neq 0$ write $e = v(z) + 2v(D)$. If $v(z)$ is odd, then e is odd and in particular is not zero. Thus $v(1 + zD^2) = \min(0, e)$, and subtraction gives the first formula. For $D = 0$ the assertions follow directly.

Suppose now that $v(z) = 0$. If $v(D) < 0$, the same computation gives valuation zero. If $v(D) > 0$, the denominator is a unit and the valuation is $2v(D) > 0$. Finally, if $v(D) = 0$, the numerator is a unit and the denominator has nonnegative valuation, so the quotient has nonpositive valuation. This proves the equivalence, including the zero value of D. □

In particular, if $v(z) = \nu > 0$ is odd and K is a nonnegative integer, then

$$v(z^K \Psi_z(D)) \geq K\nu \tag{1}$$

for every D. The assertion is uniform in D: a pole of arbitrarily large order does not weaken the lower bound. In contrast, multiplication by z^K alone cannot give such a bound, since $v(z^K D) = K\nu + v(D)$ is unbounded below as D varies.

2 Two collections of local conditions

Let $F/\mathbb{Q}$ be a number field and let p be unramified in F. For a place $\mathfrak{p}$ above p, normalize $v_\mathfrak{p}(p) = 1$. Let $z, w \in \mathbb{Q}^\times$, and let $C_1, \ldots, C_a, S_1, \ldots, S_b$ be elements of F. Fix positive integers m_i, n_j, and put

$$A_i = 2m_i + 1, \qquad B_j = 2n_j + 1.$$

Choose an integer

$$K \geq 1 + \max\{A_1, \ldots, A_a, B_1, \ldots, B_b\}.$$

Empty collections can simply be omitted. Define

$$R = zw, \qquad E_i = z^K \Psi_z(C_i), \qquad G_j = w^K \Psi_w(S_j),$$

and

$$W = \prod_{i=1}^{a} \left(1 + R\left(\frac{E_i}{R^{A_i}}\right)^2\right) \prod_{j=1}^{b} \left(1 + R\left(\frac{G_j}{R^{B_j}}\right)^2\right). \tag{2}$$

Assume all displayed denominators are nonzero, $W \neq 0$, and

$$T = \mathrm{Tr}_{F/\mathbb{Q}}(W^{-1}) \neq 0, \qquad \mathfrak{b} = R/T^2.$$

The only property of a pole certificate needed in the following theorem is the stated exclusion of negative odd valuations of $\mathfrak{b}$.

Theorem 2.1. *Assume $v_p(\mathfrak{b})$ is not negative and odd.*

1. *If $v_p(z) = \nu > 0$ is odd and $v_p(w) = 0$, there is a place $\mathfrak{p} \mid p$ such that*

$$v_\mathfrak{p}(S_j) \geq n_j\nu + 1 \qquad (1 \leq j \leq b).$$

   At every place above p, each factor indexed by a C_i in (2) is a unit.

2. *If $v_p(w) = \nu > 0$ is odd and $v_p(z) = 0$, there is a place $\mathfrak{p} \mid p$ such that*

$$v_\mathfrak{p}(C_i) \geq m_i\nu + 1 \qquad (1 \leq i \leq a).$$

   At every place above p, each factor indexed by an S_j is a unit.

The conclusion holds without any bound on the valuations of the collection whose factors are units.

Proof. We prove the first assertion; the second follows by interchanging the two collections and z, w. Since $v_p(R) = \nu$ is odd, the valuation of every factor in (2) is the minimum of zero and the valuation of its second term. There is no cancellation between terms of equal valuation: the second term has odd valuation. Every factor therefore has nonpositive valuation, and so does W.

By (1), $v_{\mathfrak{p}}(E_i) \geq K\nu$. Consequently

$$v_{\mathfrak{p}}\left(R(E_i/R^{A_i})^2\right) \geq (1 + 2K - 2A_i)\nu > 0.$$

Thus all factors in the first product are units.

Put $q = \max_{\mathfrak{p}|p} v_{\mathfrak{p}}(W) \leq 0$. Local traces of integral elements are integral, so the decomposition of the global trace into local traces gives $v_p(T) \geq -q$. The integer

$$v_p(\mathfrak{b}) = \nu - 2v_p(T)$$

is odd. By the assumed pole exclusion it is at least one, and hence $q > -\nu/2$. Choose a place at which this maximum is attained. Since the factor valuations are nonpositive and have sum q, each one is greater than $-\nu/2$. For a factor indexed by j this gives

$$2v_{\mathfrak{p}}(G_j) - (2B_j - 1)\nu > -\nu/2,$$

or

$$v_{\mathfrak{p}}(G_j) > (B_j - 3/4)\nu > 0.$$

Here w is a unit. Lemma 1.1 shows that $v_{\mathfrak{p}}(S_j) > 0$ and $v_{\mathfrak{p}}(G_j) = 2v_{\mathfrak{p}}(S_j)$, unless $S_j = 0$, in which case the conclusion already holds. Therefore

$$v_{\mathfrak{p}}(S_j) > (n_j + 1/8)\nu > n_j\nu.$$

The valuation is an integer, proving the assertion. □

The trace argument does not require p to split completely. It uses only unramifiedness, which ensures that the odd/even valuation comparison is valid at every place. Nor does it assume that the local traces are units: cancellation in a trace only strengthens the inequality used in the proof.

2.1 Even valuations of the other coefficient

The case where one coefficient is a unit does not describe every prime occurring in a product of contact functions. The following estimate also treats a nonzero even valuation of the other coefficient.

Proposition 2.2. *Fix a positive integer m and an integer $K > 2m + 9/2$, and set $A = K - 1$. Use*

$$E_i = z^K \Psi_z(C_i), \qquad G_j = w^K \Psi_w(S_j), \qquad R = zw,$$

with denominator exponent A for every factor in (2). Suppose $v_p(w) = \nu > 0$ is odd and $v_p(z) = \mu$ is even. Retain all domain conditions and the pole exclusion of Theorem 2.1.

1. *If $0 \leq \mu \leq \nu$, there is a place $\mathfrak{p} \mid p$ such that $v_{\mathfrak{p}}(C_i) > m\nu$ for every i.*
2. *If $\mu > \nu$, there is a place $\mathfrak{p} \mid p$ such that $v_{\mathfrak{p}}(S_j) > m\mu$ for every j.*
3. *If $\mu < 0$ and $\mu + \nu < 0$, the pole exclusion is impossible. If $\mu < 0$ and $\mu + \nu > 0$, the conclusion of the first assertion holds.*

In particular, zero cannot occur as $\mu + \nu$ under these parity hypotheses.

Proof. Put $\rho = \mu + \nu$, which is odd. The valuation of each factor is the minimum of zero and the odd valuation of its second term. Thus every factor has nonpositive valuation. If $\rho < 0$, the trace bound used above gives $v_p(\mathfrak{b}) \le \rho < 0$, an odd valuation, contradicting the pole exclusion. We may therefore assume $\rho > 0$. The same trace argument gives one place at which every normalized test E_i, G_j has valuation greater than

$$(K - 7/4)(\mu + \nu). \tag{3}$$

For any D whose denominator in $\Psi_z(D)$ is nonzero, one has

$$v(\Psi_z(D)) \le \max\{0, \mu + 2v(D)\}. \tag{4}$$

Indeed the formula is an equality unless $\mu + 2v(D) = 0$; in that case cancellation in the denominator can only decrease the valuation. Zero tests satisfy every desired lower bound and may be omitted from this argument.

If $0 \le \mu \le \nu$ and $v(C_i) \le m\nu$, then

$$v(E_i) \le (K+1)\mu + 2m\nu.$$

Subtracting this upper bound from (3) gives

$$(K - 2m - 7/4)\nu - \tfrac{11}{4}\mu \ge (K - 2m - 9/2)\nu > 0,$$

a contradiction. The same computation, with μ, ν exchanged, excludes $v(S_j) \le m\mu$ when $\mu > \nu > 0$.

Finally suppose $\mu < 0 < \mu + \nu$. If $v(E_i) > 0$, then $K\mu < 0$ and (4) force $\mu + 2v(C_i) > 0$. There is then no cancellation, and $v(E_i) = (K+1)\mu + 2v(C_i)$. The strict lower bound (3) implies

$$2v(C_i) > (K - 7/4)\nu - \tfrac{11}{4}\mu > 2m\nu.$$

This proves the last assertion. □

2.2 A fixed choice of exponents

The integrality of valuations permits smaller exponents when only a fixed fourth power of each contact prime is needed, independently of the contact valuation itself.

Corollary 2.3. *Put $R = zw$, $E_i = z^7\Psi_z(C_i)$ and $G_j = w^7\Psi_w(S_j)$. Use denominator exponent seven for every factor in (2), and retain its domain and pole-exclusion assumptions.*

1. *Suppose $v_p(w) = \nu > 0$ is odd and $v_p(z) = \mu$ is even. If $\mu + \nu < 0$, there is no solution satisfying the pole exclusion. If $\mu + \nu > 0$, one of the following holds at one common place $\mathfrak{p} \mid p$:*

   (a) *if $\mu \le \nu$, then $v_\mathfrak{p}(C_i) \ge 4$ for every i;*

   (b) *if $\mu > \nu$, then $v_\mathfrak{p}(S_j) \ge 6$ for every j.*

2. *Suppose $v_p(z) = \mu > 0$ is odd and $v_p(w) = 0$. Then at one common place*

$$v_\mathfrak{p}(S_j) > \frac{25}{8}\mu, \qquad v_\mathfrak{p}(S_j) \ge 3\mu + 1 \ge 4$$

   for every j. Each factor indexed by a C_i is a unit.

The first assertion also holds when the E_i in the product are divided by a rational number δ with $v_p(\delta) = 0$. The second also holds with this division if $v_p(\delta) \le 0$.

Proof. At a prime where $\rho = v_p(R)$ is odd, all factor valuations are nonpositive. The trace argument proves that $\rho > 0$ is necessary and supplies a place at which each normalized test has valuation strictly greater than $25\rho/4$.

Suppose first that $\mu \geq 0$ is even and $\mu \leq \nu$. For a comparison test C_i this lower bound forces $\mu + 2v(C_i) > 0$: otherwise $v(E_i) \leq 7\mu$, whereas $25(\mu+\nu)/4 > 7\mu$. Consequently

$$v(C_i) > \frac{25\nu - 7\mu}{8}.$$

If $\nu = 1$, parity gives $\mu = 0$, so $v(C_i) > 25/8$ and the integer valuation is at least four. If $\nu \geq 3$, the bound is at least $18\nu/8 \geq 54/8 > 6$, which is stronger. For $\mu > \nu > 0$, the symmetric argument gives

$$v(S_j) > \frac{25\mu - 7\nu}{8} \geq \frac{18\mu + 7}{8} \geq \frac{43}{8}.$$

Here $\mu \geq 2$ is even and $\nu \leq \mu - 1$, so the valuation is at least six. For $\mu < 0 < \mu + \nu$, positivity of the extracted E_i forces $\mu + 2v(C_i) > 0$ and gives the same displayed comparison bound, now greater than $25\nu/8$, hence at least four.

In the second assertion, Lemma 1.1 gives $v(E_i) \geq 7\mu$. The second term of the corresponding factor therefore has valuation at least $\mu > 0$. Those factors are units. For the remaining factors the trace estimate gives $v(G_j) > 25\mu/4$. Since w is a unit, Lemma 1.1 gives $v(G_j) = 2v(S_j)$, or $S_j = 0$. This proves the stated bound; integrality and $25\mu/8 > 3\mu$ give $v(S_j) \geq 3\mu + 1$. Finally, dividing E_i by a unit has no effect. Dividing it by δ of nonpositive valuation only increases its valuation, so its factors remain units in the second assertion. □

3 A unit among two balanced contact functions

Let Θ_0, Θ_1 be finite sets of nonzero rational numbers, and let $c_0, c_1 \in \mathbb{Q}^\times$ satisfy $c_i \notin \Theta_i$. Assume

$$(\Theta_0 \cup \{c_0\}) \cap (\Theta_1 \cup \{c_1\}) = \varnothing.$$

Put $d_i = |\Theta_i|$ and define the balanced rational functions

$$R_i(t) = \frac{\prod_{\theta \in \Theta_i}(t - \theta)}{(t - c_i)^{d_i}}.$$

The term balanced here means precisely that numerator and denominator have the same degree; no additional property is included in the term. If the intended application needs a square denominator, take d_i even. Let S contain all prime divisors of the numerators and denominators of the nonzero members of the two displayed sets and of their nonzero pairwise differences.

Theorem 3.1. *Let $r, T, \lambda \in \mathbb{Q}$, with $r \neq 0$, and set*

$$a = (1 + rT^2)^2, \qquad t_0 = \lambda, \quad t_1 = a\lambda.$$

Suppose $p \notin S$ and $v_p(r)$ is odd. If $R_i(t_i)$ are defined and nonzero, then at least one is a p-adic unit:

$$v_p(R_0(t_0)) = 0 \quad \text{or} \quad v_p(R_1(t_1)) = 0.$$

The conclusion permits negative as well as positive odd $v_p(r)$.

Proof. If $T = 0$, then $a = 1$. Otherwise $v_p(rT^2)$ is odd and cannot be zero. If it is positive, a is a unit congruent to one modulo p. If it is negative, $v_p(a) = 2v_p(rT^2) < 0$. Thus in every case $a \neq 0$ and either $a \equiv 1 \pmod{p}$ or $v_p(a) < 0$.

For either balanced contact function, an argument of negative valuation has a numerator and denominator of the same valuation $d_i v_p(t)$, so its value is a unit. An argument of positive valuation, including zero, also gives a unit, since all the contacts and the pole are units.

It remains to consider $v_p(\lambda) = 0$ and $v_p(R_0(\lambda)) \neq 0$. The residue of λ must then belong to $\Theta_0 \cup \{c_0\}$. If $v_p(a) < 0$, the argument $a\lambda$ has negative valuation, and $R_1(a\lambda)$ is a unit. Otherwise $a\lambda \equiv \lambda \pmod{p}$. The two contact-and-pole sets have disjoint reductions modulo p by the choice of S, so this residue does not belong to $\Theta_1 \cup \{c_1\}$. Every numerator and denominator factor of $R_1(a\lambda)$ is then a unit. □

4 Scope for Diophantine constructions

Theorem 3.1 supplies one contact function of valuation exactly zero at a prime of odd valuation of r. Thus Theorem 2.1 can extract scalar local conditions from at least one product $rR_i(t_i)$, while its other collection of local conditions is suppressed independently of any pole depths. At a comparison prime where r is a unit and $R_i(t_i)$ has positive odd valuation, the same theorem instead extracts the comparison conditions. Every transformation is rational and introduces no new unrestricted rational parameter; its denominators must be retained as nonzero guards in an algebraic presentation.

For a single product, two odd coefficient valuations have even sum, so its pole condition alone need not extract either collection. In the two-contact-function construction, Theorem 3.1 then supplies the other function with valuation zero and permits scalar extraction there. Together with the even-valuation cases, this gives the alternative needed in the companion twenty-five-variable construction: each comparison prime divides either an integer residual or the scalar index to the required order. The global height inequalities and simultaneous rational realization are proved separately in that construction. The local lemmas here do not establish its inherited arithmetic premises.

Exact finite verification

The standard-library program

`computation/local_tests/check_rational_normalization.py`

checks the rational-function formulas, including equal-valuation cancellation, over explicit finite ranges; it separately verifies the linear inequalities in Proposition 2.2 and the integer rounding in Corollary 2.3. The certificate records 38 725 rational calculations, 60 501 cases of each precision argument, and 4 680 examples of Theorem 3.1. In the exponent-seven specialization, the comparison order is at least four and an odd scalar coefficient order $\mu > 0$ gives scalar order at least $3\mu + 1$. The proofs in this paper establish the unrestricted statements; these finite checks verify arithmetic and specializations used in the accompanying construction.